Shehtab Zaman

New York, NY · 917-651-3696 · shehtabzaman@gmail.com · github.com/szaman19 · linkedin.com/in/shehtabzaman

Research scientist building AI systems that accelerate scientific discovery across medicine, chemistry, and materials. Currently developing digital twins and world models for cancer radiotherapy at Memorial Sloan Kettering Cancer Center from multimodal patient data (CT, PET, MRI, clinical records), and the creator of **Charge**, a multi-agent chemical reasoning framework in use at LLNL. Six years of distributed deep learning at 100+ GPU scale, building custom CUDA/NVSHMEM kernels, generative models, and scientific foundation models.

EXPERIENCE

Research Fellow, Department of Medical Physics 2026 – Present

Memorial Sloan Kettering Cancer Center · New York, NY

- Developing patient-specific **digital twins and world models for cancer radiotherapy**, learning forward dynamics of tumor and normal-tissue response to inform adaptive treatment planning
- Building multimodal representation learning over CT, PET, and MRI volumes fused with structured clinical data, spanning registration, segmentation, and longitudinal change modeling
- Pretraining 3D vision foundation models on large retrospective imaging cohorts, with transfer to outcome and dose-response prediction
- Designing scalable training and inference pipelines for volumetric medical data under clinical data-governance constraints

Graduate Research Assistant, Computer Science Dec 2019 – 2026

Binghamton University, SUNY · Binghamton, NY

- Designed and implemented scalable deep learning systems for training LLMs and GNNs on large-scale molecular datasets across multi-node, multi-GPU clusters
- Built distributed pre-training and fine-tuning pipelines with PyTorch, Megatron-LM, NCCL, and NVSHMEM for efficient training on HPC infrastructure
- Developed custom distributed communication backends and high-throughput data pipelines for large-scale scientific datasets
- Led a team building end-to-end ML systems for materials and drug discovery; deployed generative AI pipelines with TotalEnergies and LLNL
- Co-authored a **$10M funded research initiative** with LLNL advancing AI-driven scientific discovery

Computing Scholar Intern Summer 2022

Lawrence Livermore National Laboratory · Livermore, CA

- Implemented tensor-parallel distributed training for graph neural networks in LBANN, scaling to large graph workloads across multi-GPU systems
- Wrote GPU kernels with GPU-initiated partitioned global memory access for Gather-Scatter operations using NVIDIA NVSHMEM; profiled multi-GPU CUDA kernels with NSight

Computing Scholar Intern Summer 2021

Lawrence Livermore National Laboratory · Livermore, CA

- Built custom CUDA kernels adding GNN support to a large-scale training toolkit, scaling to **100+ GPUs** for 12×–93× speedup over PyTorch baselines and inference on 3M samples in under 30 seconds

Data Science Institute Research Scholar, Lawrence Livermore National Laboratory — distributed GPU-accelerated graph convolutional networks in the LBANN toolkit Summer 2020

EDUCATION

Ph.D., Computer Science — Binghamton University, SUNY 2021 – 2026

M.S., Computer Science — Binghamton University, SUNY · 3.9/4.0 2019 – 2020

B.S., Mathematical Physics, minor in Computer Science — Binghamton University · 3.9/4.0 2015 – 2019

TECHNICAL SKILLS

Languages	Python, C++, CUDA
ML	PyTorch, TensorFlow, Hugging Face, Megatron-LM, LBANN, scikit-learn
Domains	Computer vision, volumetric medical imaging, foundation models, generative & diffusion models, gnns, NLP
Agents	Multi-agent frameworks, MCP tool servers, LLM post-training & evaluation, OpenAI / Gemini / Ollama / vLLM
HPC	NCCL, NVSHMEM, MPI, multi-node GPU training, NSight profiling
Data	CT / PET / MRI, DICOM, clinical records, molecular and materials datasets

SELECTED PROJECTS & LIBRARIES

Charge — agentic chemical reasoning framework · creator

- Multi-agent framework for chemistry problem solving — lead molecule optimization, reaction prediction, and structured tool use
- In active use at LLNL for agentic scientific tooling and showcased at OpenAI events
- Automatic MCP server generation for rapid tool creation, with structured agent outputs that drop straight into downstream applications
- Provider-agnostic across OpenAI, Gemini, and local backends (Ollama, vLLM)

DGraph — distributed dense graph neural networks

- Supports arbitrary GNN architectures, including heterogeneous, multi-graph, and edge-conditioned networks
- Written against NCCL, MPI, and NVSHMEM for portability across communication libraries, with first-class GPU-initiated one-sided communication; tested with full-graph training on multi-billion-vertex graphs

ParticleGrid — high-performance C++ library for 3D molecular representation

- SIMD-accelerated generation of 3D molecular and crystal representations, up to 10,000× faster than NumPy equivalents
- Python interface with zero-copy interoperability across PyTorch, NumPy, and TensorFlow; built with TotalEnergies and released as a production-quality library — github.com/particlegrid/particlegrid

SELECTED PUBLICATIONS

BOOM: Benchmarking Out-of-distribution Molecular Property Prediction *NeurIPS 2025, D&B Track*
E. Antoniuk*, **S. Zaman***, T. Ben-Nun, P. Li, J. Diffenderfer, et al.

Score-based Idempotent Distillation of Diffusion Models *SPIGM Workshop, NeurIPS 2025*
S. Zaman, C. Liu, K. Chiu

PtychoFormer: A Transformer-based Model for Ptychographic Phase Retrieval *Under review, 2024*
R. Nakahata, **S. Zaman**, M. Zhang, F. Lu, K. Chiu

STRIDE: Structure-guided Generation for Inverse Design of Molecules *AI4Science Workshop, NeurIPS 2023*
S. Zaman, D. Akhiyarov, M. Araya-Polo, K. Chiu

ParticleGrid: Enabling Deep Learning using 3D Representation of Materials *IEEE eScience 2022*
E. Ferguson*, **S. Zaman***, C. Pereira, D. Akhiyarov, M. Araya-Polo, K. Chiu

Parallelizing Graph Neural Networks via Matrix Compaction *IEEE CCGrid 2022*
S. Zaman, T. Moon, T. Benson, S. A. Jacobs, K. Chiu, B. Van Essen

MoleculeFlow: A Deep Generative Workflow for 3D Molecular Generation *ML4Molecules Workshop, NeurIPS 2021*
E. Ferguson*, **S. Zaman***, M. Araya-Polo, D. Akhiyarov, K. Chiu, C. Pereira

Graph Neural Network for MOF Potential Energy Approximation *ML for Molecules Workshop, NeurIPS 2020*
S. Zaman, C. Owen, K. Chiu, M. Lawler

Real-Space Visualization of Quantum Phase Transition by Network Topology *Physical Review E 100(1), 2019*
S. Zaman, W. C. Lee